

深圳市英瑞达光电有限公司

Specification for 4.26inch EPD

Model NO. : YRD0426BBS770F0

Confirmation:

Prepared by	Checked by	Approved by

Customer approval:

Customer	Approved by	Date

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1. Over View

DEPG0426BBS770F0 is an Active Matrix Electrophoretic Display (AM EPD), with interface and a reference system design. The display is capable to display image at 1-bit white and black full display capabilities. The 4.26 inch active area contains 800×480 pixels. The module is a TFT-array driving electrophoresis display, with integrated circuits including gate driver, source driver, MCU interface, timing controller, oscillator, DC-DC, SRAM, LUT, VCOM. Module can be used in portable electronic devices, such as Electronic Shelf Label (ESL) System.

2. Features

- ◆ 800×480 pixels display
 - ◆ High contrast High reflectance
 - ◆ Ultra wide viewing angle Ultra low power consumption
 - ◆ Pure reflective mode
 - ◆ Bi-stable display
 - ◆ Commercial temperature range
 - ◆ Landscape portrait modes
 - ◆ Hard-coat antiglare display surface
 - ◆ Ultra Low current deep sleep mode
 - ◆ On chip display RAM
 - ◆ Waveform can stored in On-chip OTP or written by MCU
 - ◆ Serial peripheral interface available
 - ◆ On-chip oscillator
 - ◆ On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
 - ◆ Built-in temperature sensor
-

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3. Mechanical and Optical Specification

Parameter	Specifications	Unit	Remark
Screen Size	4.26	Inch	
Display Resolution	480(H)×800(V)	Pixel	DPI:219
Active Area	55.68(H)×92.80(V)	mm	
Pixel Pitch	0.116×0.116	mm	
Pixel Configuration	Rectangle		
Outline Dimension	62.37(H)×105.33 (V) ×1.0±0.10	mm	
Weight	±0.5	g	

Symbol	Parameter	Conditions	Min	Typ.	Max	Units	Notes
KS	Black State L* value		-	24	30		3-1
WS	White State L* value		58	62	-		3-1
	White Ghosting ΔL	Full Display Mode	-	1	-		3-1
		Partil Display Mode	-	1	-		3-1
		Full-Partil Display Mode	-	1	-		3-1
R	White Reflectivity	White	-	30	-	%	3-1
CR	Contrast Ratio	Indoor	-	7	-		3-1 3-2
GN	2Grey Level	-	-	-	-		
Life		Temp:23±3°C Humidity:55±10%RH		5years			3-3

Notes: 3-1. Luminance meter: Eye-One Pro Spectrophotometer.

3-2. CR=Surface Reflectance with all white pixel/Surface Reflectance with all black pixels.

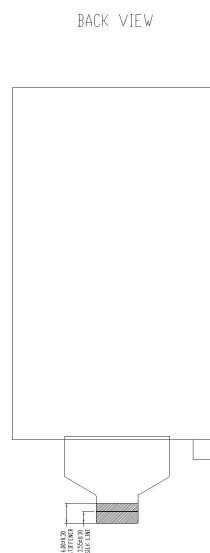
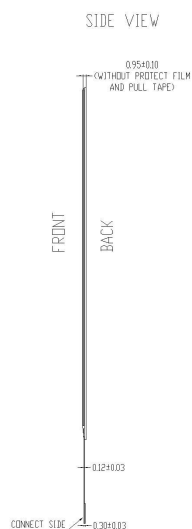
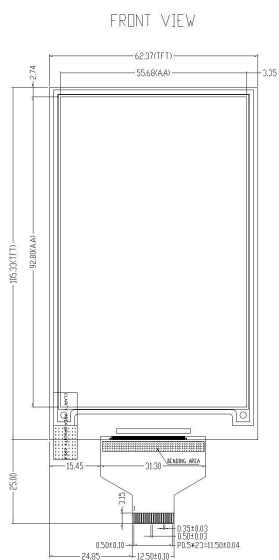
3-3. When the product is stored. The display screen should be kept white and face up.

4.Mechanical Drawing of EPD Module

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Confirmation:

DATE	REV.	MODIFICATION
2025-11-20	A0	FIRST ISSUE
2025-12-03	B0	ADD FPC LENGTH



NOTE

- 1 DISPLAY MODULE 4.26" ARRAY FOR EPD
2 DRIVER IC:SSD1677
3 RESOLUTION:800gateX480source
4 PIXEL SIZE: 0.116mmX 0.116mm
5 FOR GY

5. Input/output Pin Assignment

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No.	Name	I/O	Description	Remark
1	NC		Do not connect with other NC pins	Keep Open
2	GDR	O	N-Channel MOSFET Gate Drive Control	
3	RESE	I	Current Sense Input for the Control Loop	
4	NC		Do not connect with other NC pins	Keep Open
5	VSH2	C	Positive Source driving voltage	
6	NC		Do not connect with other NC pins	Keep Open
7	NC		Do not connect with other NC pins	Keep Open
8	BS1	I	Bus Interface selection pin	Note 5-5
9	BUSY	O	Busy state output pin	Note 5-4
10	RES#	I	Reset signal input. Active Low.	Note 5-3
11	D/C#	I	Data /Command control pin	Note 5-2
12	CS#	I	Chip select input pin	Note 5-1
13	SCL	I	Serial Clock pin (SPI)	
14	SDA	I/O	Serial Data pin (SPI)	
15	VDDIO	P	Power Supply for interface logic pins It should be connected with VCI	
16	VCI	P	Power Supply for the chip	
17	VSS	P	Ground	
18	VDD	C	Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS	
19	VPP	P	FOR TEST	Keep Open
20	VSH1	C	Positive Source driving voltage	
21	VGH	C	Power Supply pin for Positive Gate driving voltage and VSH1	
22	VSL	C	Negative Source driving voltage	
23	VGL	C	Power Supply pin for Negative Gate driving voltage VCOM and VSL	
24	VCOM	C	VCOM driving voltage	

I = Input Pin, O =Output Pin,I /O = Bi-directional Pin (Input/output), P = Power Pin, C = Capacitor Pin

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Note 5-1: This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CS# is pulled LOW.

Note 5-2: This pin is (D/C#) Data/Command control pin connecting to the MCU in 4-wire SPI mode. When the pin is pulled HIGH, the data at SDA will be interpreted as data. When the pin is pulled LOW, the data at SDA will be interpreted as command.

Note 5-3: This pin (RES#) is reset signal input. The Reset is active low.

Note 5-4: This pin is Busy state output pin. When Busy is High, the operation of chip should not be interrupted, command should not be sent. The chip would put Busy pin High when -Outputting display waveform -Communicating with digital temperature sensor.

Note 5-5: Bus interface selection pin.

BS1 State	MCU Interface
L	4-lines serial peripheral interface(SPI) - 8 bits SPI
H	3- lines serial peripheral interface(SPI) - 9 bits SPI

6. Electrical Characteristics

6.1 Absolute Maximum Rating

Parameter	Symbol	Rating	Unit
Logic supply voltage	VCI,VDDIO	-0.5 to +4.0	V
Logic Input voltage	VIN	-0.5 to V _{DDIO} +0.5	V
Logic Output voltage	VOUT	-0.5 to V _{DDIO} +0.5	V
Operating Temp range	TOPR	0 to +50	°C.
Storage Temp range	TSTG	-25 to+70	°C.
Optimal Storage Temp	TSTGo	23±3	°C.
Optimal Storage Humidity	HSTGo	55±10	%RH

Note:

- 1. Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables.
- 2. The display screen should be kept white and face up during storage. Please refer to the [Reliability Test] section.

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6.2 Panel DC Characteristics

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23°C.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
Single ground	V _{SS}	-		-	0	-	V
Logic supply voltage	V _{CI}	-	V _{CI}	2.2	3.0	3.3	V
Power for interface logic pins	V _{DDIO}		V _{DDIO}	2.2	-	3.3	V
Core logic voltage	V _{DD}		V _{DD}	1.7	1.8	1.9	V
VCOM_DC output voltage	V _{com}	-	V _{COM}	-4.0	-	-0.1	V
High level input voltage	V _{IH}	-	-	0.8 V _{DDIO}	-	-	V
Low level input voltage	V _{IL}	-	-	-	-	0.2 V _{DDIO}	V
High level output voltage	V _{OH}	IOH = -100uA	-	0.9 V _{DDIO}	-	-	V
Low level output voltage	V _{OL}	IOL = 100uA	-	-	-	0.1 V _{DDIO}	V
	Typical power	P _{TYP}	V _{CI} =3.0V	-	-	25.5	- mW
	Deep sleep mode	P _{STPY}	V _{CI} =3.0V	-	-	0.0003	- mW
	Typical operating current	Iopr_VCI	V _{CI} =3.0V	-	-	8.5	- mA
	Image update time	-	23 °C	-	-	1.1	- sec
	Typical peak current	Iopr_VCI	2.3~3.6v			85.5	- mA
Deep sleep mode current	Idslp_VCI	DC/DC off No clock No input load Ram data not retain	-	-	1	5	uA

Notes: 1. The typical power measurement method is as follows, and the output is reported according to the test screen provided by the YRD.



- 2. The deep sleep power is the consumed power when the panel controller is in deep sleep mode.
- 3. The listed electrical characteristics are only guaranteed under the controller & waveform provided by YRD.
- 4. Electrical measurement: Tektronix oscilloscope - MSO44,
Tektronix current probe - TCPA300.

6.3 Panel AC Characteristics

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23°C.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
VCOM output voltage	VCOM	VCI=3.0V Enable Clock and Analog by Master Activation Command VGH=20V VGL=-VGH VSH1=15V VSH2=5V VSL=-15V VCOM = -2V No waveform transitions. No loading. No RAM read/write No OTP read /write	VCOM	-2.2	-2	-1.8	V
Positive Source output voltage	VSH		S0~S127	+14.8	+15	+15.2	V
Negative Source output voltage	VSL		S0~S127	-15.2	-15	-14.8	V
Positive gate output voltage	Vgh		G0~G295	+19.5	+20	+20.5	V
Negative gate output voltage	Vgl		G0~G295	-20.5	-20	-19.5	V

6.4 MCU Interface

6.4.1 MCU Interface Selection

The pin assignment at different interface mode is summarized in Table 6-4-1. Different MCU mode can be set by hardware selection on BS1 pins. The display panel only supports 4-wire SPI or 3-wire SPI interface mode.

Pin Name	Data/Command Interface		Control Signal		
Bus interface	SDA	SCL	CS#	D/C#	RES#
BS1=L 4-wire SPI	SDA	SCL	CS#	D/C#	RES#
BS1=H 3-wire SPI	SDA	SCL	CS#	L	RES#

Table 6-4-1: MCU interface assignment under different bus interface mode

6.4.2 MCU Serial Interface (4-wire SPI)

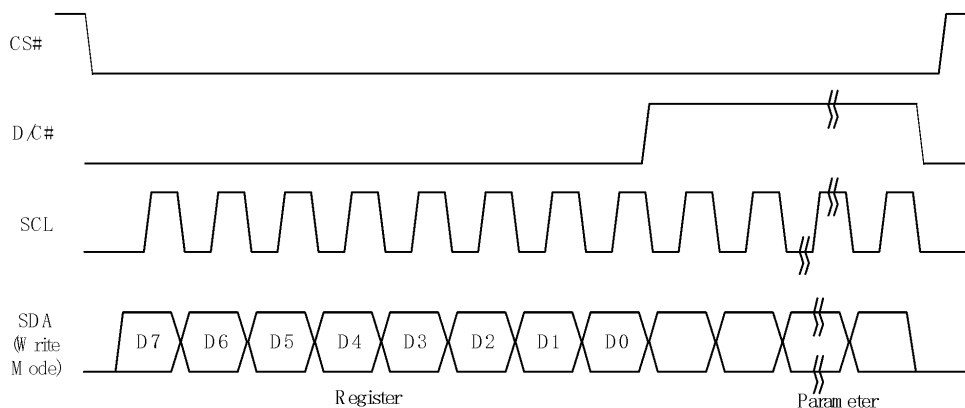
The serial interface consists of serial clock SCL, serial data SDA, D/C#, CS#. This interface supports Write mode and Read mode.

Function	CS#	D/C#	SCL
Write command	L	L	↑
Write data	L	H	↑

Note: ↑ stands for rising edge of signal

In the write mode SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C# should be kept over the whole byte . The data byte in the shift register is written to the Graphic Display Data RAM /Data Byte register or command Byte register according to D/C# pin.

Figure 6-1: Write procedure in 4-wire SPI mode

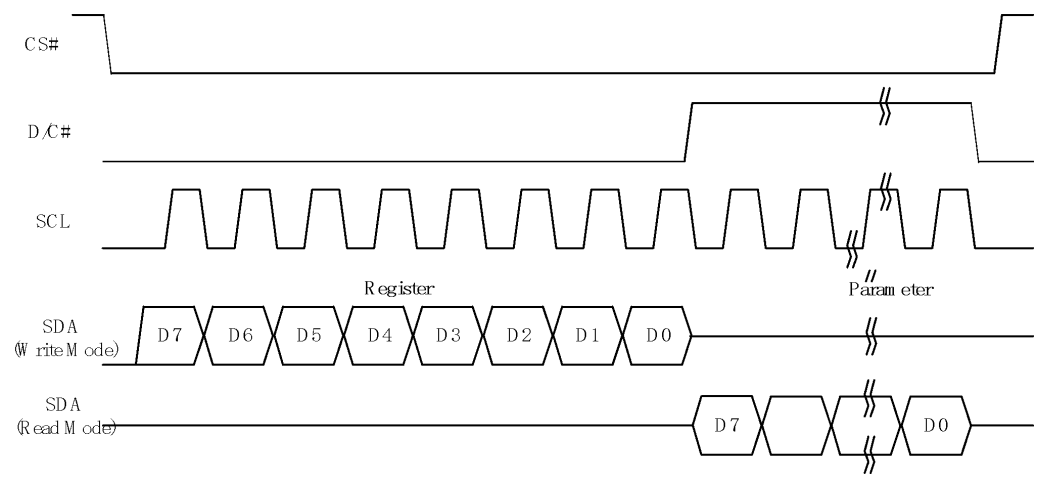


In the Read mode:

1. After driving CS# to low, MCU need to define the register to be read.
2. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0 with D/C# keep low.
3. After SCL change to low for the last bit of register, D/C# need to drive to high.
4. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
5. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

Figure 6-2: Read procedure in 4-wire SPI mode

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6.4.3 MCU Serial Interface (3-wire SPI)

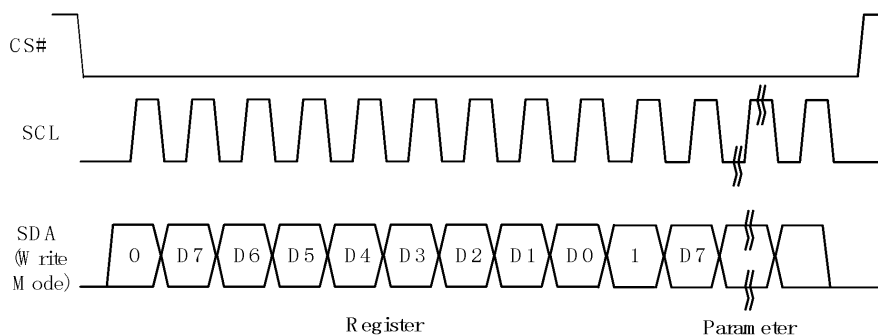
The 3-wire serial interface consists of serial clock SCL, serial data SDA and CS#. This interface also supports Write mode and Read mode.

The operation is similar to 4-wire serial interface while D/C# pin is not used. There are altogether 9-bits will be shifted into the shift register on every ninth clock in sequence: D/C# bit, D7 to D0 bit. The D/C# bit (first bit of the sequential data) will determine the following data byte in the shift register is written to the Display Data RAM (D/C# bit = 1) or the command register (D/C# bit = 0).

Function	CS#	D/C#	SCL
Write command	L	Tie	↑
Write data	L	Tie	↑

Note: ↑ stands for rising edge of signal

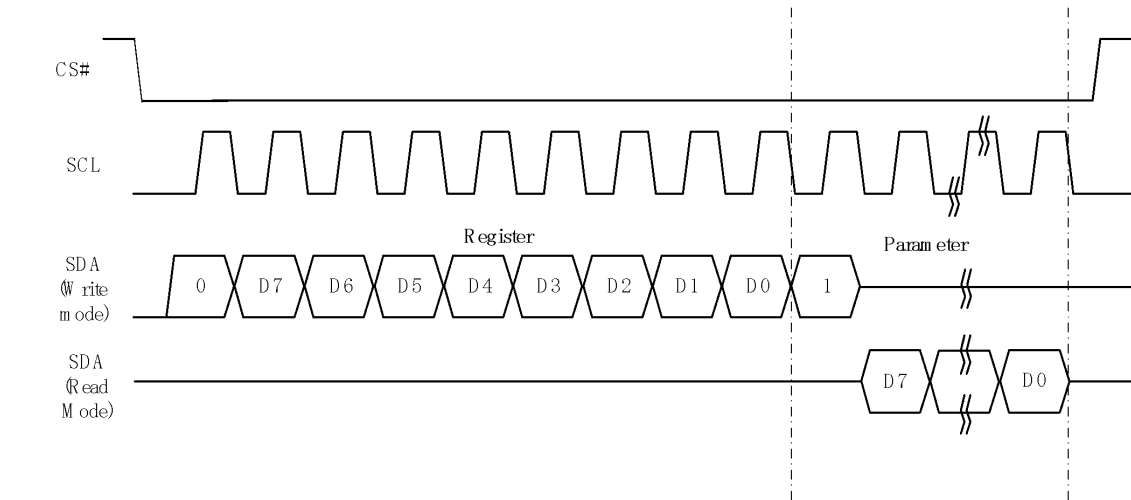
Figure 6-3: Write procedure in 3-wire SPI mode



In the Read mode:

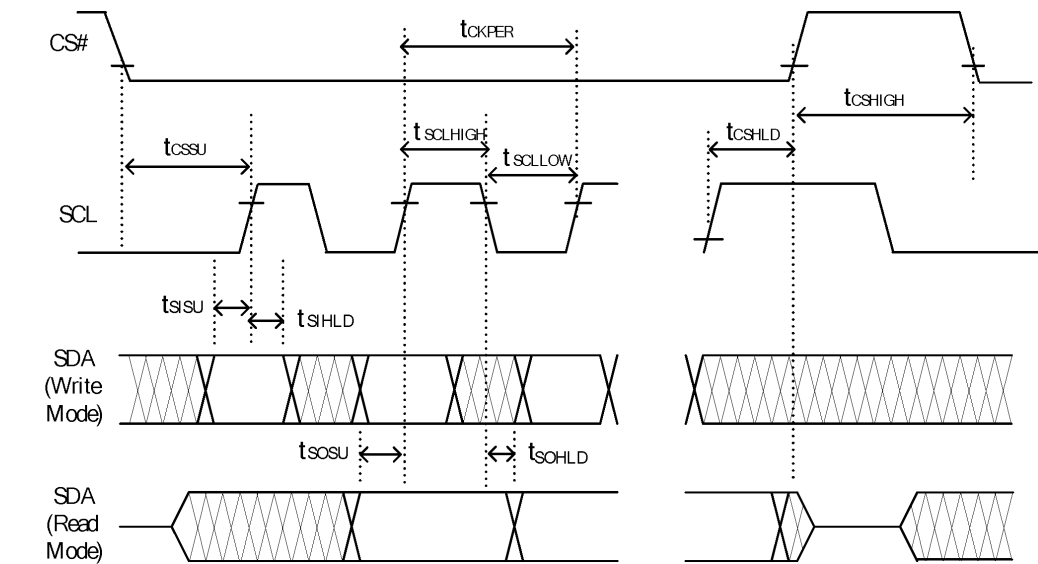
1. After driving CS# to low, MCU need to define the register to be read.
2. D/C=0 is shifted thru SDA with one rising edge of SCL
3. SDA is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0.
4. D/C=1 is shifted thru SDA with one rising edge of SCL
5. SDA is shifted out an 8-bit data on every falling edge of SCL in the order of D7, D6, ... D0.
6. Depending on register type, more than 1 byte can be read out. After all byte are read, CS# need to drive to high to stop the read operation.

Figure 6-4: Read procedure in 3-wire SPI mode



6.4.4 Interface Timing

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23°C.



Changed Diagram

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Serial Interface Timing Characteristics

(VDDIO - VSS = 2.2V to 3.7V, TOPR = 23°C, CL=30pF)

Write mode

Symbol	Parameter	Min	Typ.	Max	Unit
fSCL	SCL frequency (Write Mode)			20	MHz
tCSSU	Time CS# has to be low before the first rising edge of SCLK	20			ns
tCSHLD	Time CS# has to remain low after the last falling edge of SCLK	20			ns
tCSHIGH	Time CS# has to remain high between two transfers	100			ns
tSCLCYC	SCL cycle time	50			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	25			ns
tSCLLOW	Part of the clock period where SCL has to remain low	25			ns
tSISU	Time SI (SDA Write Mode) has to be stable before the next rising edge of SCL	10			ns
tSIHLD	Time SI (SDA Write Mode) has to remain stable after the rising edge of SCL	40			ns

Read mode

Symbol	Parameter	Min	Typ.	Max	Unit
fSCL	SCL frequency (Read Mode)			2.5	MHz
tCSSU	Time CS# has to be low before the first rising edge of SCLK	100			ns
tCSHLD	Time CS# has to remain low after the last falling edge of SCLK	50			ns
tCSHIGH	Time CS# has to remain high between two transfers	250			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	180			ns
tSCLLOW	Part of the clock period where SCL has to remain low	180			ns
tSOSU	Time SO(SDA Read Mode) will be stable before the next rising edge of SCL		50		ns
tSOHLD	Time SO (SDA Read Mode) will remain stable after the falling edge of SCL		0		ns

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7.Command Table

Command Table											Command	Description																																													
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Driver Output control	Gate setting A[9:0]= 2A7h [POR], 680 MUX MUX Gate lines setting as (A[9:0] + 1). B[2:0] = 000 [POR]. Gate scanning sequence and direction B[2]: GD Selects the 1st output Gate GD=0 [POR]. G0 is the 1st gate output channel, gate output sequence is G0,G1, G2, G3, ... GD=1, G1 is the 1st gate output channel, gate output sequence is G1, G0, G3, G2, ... B[1]: SM Change scanning order of gate driver. SM=0 [POR]. G0, G1, G2, G3...679 (left and right gate interlaced) SM=1, G0, G2, G4 ...G678, G1, G3, ...G679 B[0]: TB TB = 0, Scan from G0 to G679 TB = 1, Reserved																																													
0	0	01	0	0	0	0	0	0	0	1																																															
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																															
0	1		0	0	0	0	0	0	A ₉	A ₈																																															
0	1		0	0	0	0	0	0	B ₂	B ₁	B ₀																																														
0	0	03	0	0	0	0	0	0	1	1	Gate Driving voltage Control	Set Gate driving voltage A[4:0] = 00h [POR] VGH setting from 12V to 20V																																													
0	1		0	0	0	A ₄	A ₃	A ₂	A ₁	A ₀																																															
													<table><tr><th>A[4:0]</th><th>VGH</th><th>A[4:0]</th><th>VGH</th></tr><tr><td>00h</td><td>20</td><td>10h</td><td>16.5</td></tr><tr><td>07h</td><td>12</td><td>11h</td><td>17</td></tr><tr><td>08h</td><td>12.5</td><td>12h</td><td>17.5</td></tr><tr><td>09h</td><td>13</td><td>13h</td><td>18</td></tr><tr><td>0Ah</td><td>13.5</td><td>14h</td><td>18.5</td></tr><tr><td>0Bh</td><td>14</td><td>15h</td><td>19</td></tr><tr><td>0Ch</td><td>14.5</td><td>16h</td><td>19.5</td></tr><tr><td>0Dh</td><td>15</td><td>17h</td><td>20</td></tr><tr><td>0Eh</td><td>15.5</td><td>Other</td><td>NA</td></tr><tr><td>0Fh</td><td>16</td><td></td><td></td></tr></table>	A[4:0]	VGH	A[4:0]	VGH	00h	20	10h	16.5	07h	12	11h	17	08h	12.5	12h	17.5	09h	13	13h	18	0Ah	13.5	14h	18.5	0Bh	14	15h	19	0Ch	14.5	16h	19.5	0Dh	15	17h	20	0Eh	15.5	Other	NA	0Fh	16		
A[4:0]	VGH	A[4:0]	VGH																																																						
00h	20	10h	16.5																																																						
07h	12	11h	17																																																						
08h	12.5	12h	17.5																																																						
09h	13	13h	18																																																						
0Ah	13.5	14h	18.5																																																						
0Bh	14	15h	19																																																						
0Ch	14.5	16h	19.5																																																						
0Dh	15	17h	20																																																						
0Eh	15.5	Other	NA																																																						
0Fh	16																																																								

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Command Table

R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	04	0	0	0	0	0	1	0	0	Source Driving voltage Control	Set Source driving voltage A[7:0] = 41h [POR], VSH1 at 15V B[7:0] = A8h [POR], VSH2 at 5V. C[7:0] = 32h [POR], VSL at -15V
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		

B[7] = 1,
VSH2 voltage setting from 2.4V to 8.8V

A/B[7:0]	VSH1/VSH2	A/B[7:0]	VSH1/VSH2
8Eh	2.4	AFh	5.7
8Fh	2.5	B0h	5.8
90h	2.6	B1h	5.9
91h	2.7	B2h	6
92h	2.8	B3h	6.1
93h	2.9	B4h	6.2
94h	3	B5h	6.3
95h	3.1	B6h	6.4
96h	3.2	B7h	6.5
97h	3.3	B8h	6.6
98h	3.4	B9h	6.7
99h	3.5	BAh	6.8
9Ah	3.6	BBh	6.9
9Bh	3.7	BCh	7
9Ch	3.8	BDh	7.1
9Dh	3.9	BEh	7.2
9Eh	4	BFh	7.3
9Fh	4.1	C0h	7.4
A0h	4.2	C1h	7.5
A1h	4.3	C2h	7.6
A2h	4.4	C3h	7.7
A3h	4.5	C4h	7.8
A4h	4.6	C5h	7.9
A5h	4.7	C6h	8
A6h	4.8	C7h	8.1
A7h	4.9	C8h	8.2
A8h	5	C9h	8.3
A9h	5.1	CAh	8.4
AAh	5.2	CBh	8.5
ABh	5.3	CAh	8.6
ACH	5.4	CDh	8.7
ADh	5.5	CEh	8.8
Aeh	5.6	Other	NA

A[7]/B[7] = 0,
VSH1/VSH2 voltage setting from 9V to 17V

A/B[7:0]	VSH1/VSH2	A/B[7:0]	VSH1/VSH2
23h	9	3Ch	14
24h	9.2	3Dh	14.2
25h	9.4	3Eh	14.4
26h	9.6	3Fh	14.6
27h	9.8	40h	14.8
28h	10	41h	15
29h	10.2	42h	15.2
2Ah	10.4	43h	15.4
2Bh	10.6	44h	15.6
2Ch	10.8	45h	15.8
2Dh	11	46h	16
2Eh	11.2	47h	16.2
2Fh	11.4	48h	16.4
30h	11.6	49h	16.6
31h	11.8	4Ah	16.8
32h	12	4Bh	17
33h	12.2	Other	NA
34h	12.4		
35h	12.6		
36h	12.8		
37h	13		
38h	13.2		
39h	13.4		
3Ah	13.6		
3Bh	13.8		

C[7] = 0,
VSL setting from -9V to -17V

C[7:0]	VSL
1Ah	-9
1Ch	-9.5
1Eh	-10
20h	-10.5
22h	-11
24h	-11.5
26h	-12
28h	-12.5
2Ah	-13
2Ch	-13.5
2Eh	-14
30h	-14.5
32h	-15
34h	-15.5
36h	-16
38h	-16.5
3Ah	-17
Other	NA

Remark:
VSH1 > VSH2

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Command Table

R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																								
0	0	08	0	0	0	0	1	0	0	0	Initial Code Setting OTP Program	Program Initial Code Setting The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.																								
0	0	09	0	0	0	0	1	0	0	1	Write Register for Initial Code Setting	Write Register for Initial Code Setting Selection A[7:0] ~ D[7:0]: Reserved Details refer to Application Notes of Initial Code Setting																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																										
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																										
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀																										
0	0	0A	0	0	0	0	1	0	1	0	Read Register for Initial Code Setting	Read Register for Initial Code Setting																								
0	0	0C	0	0	0	0	1	1	0	0	Booster Soft-start Control	This command is used to control the inrush current for the booster. Two level of strengths can be selected for the booster <table><tr><th></th><th colspan="5">Data bytes</th></tr><tr><th></th><th>A[7:0]</th><th>B[7:0]</th><th>C[7:0]</th><th>D[7:0]</th><th>E[7:0]</th></tr><tr><td>Level 1</td><td>AE</td><td>C7</td><td>C3</td><td>C0</td><td>40</td></tr><tr><td>Level 2</td><td>AE</td><td>C7</td><td>C3</td><td>C0</td><td>80</td></tr></table>		Data bytes						A[7:0]	B[7:0]	C[7:0]	D[7:0]	E[7:0]	Level 1	AE	C7	C3	C0	40	Level 2	AE	C7	C3	C0	80
	Data bytes																																			
	A[7:0]	B[7:0]	C[7:0]	D[7:0]	E[7:0]																															
Level 1	AE	C7	C3	C0	40																															
Level 2	AE	C7	C3	C0	80																															
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																										
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																										
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀																										
			E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀																										
0	0	10	0	0	0	1	0	0	0	0	Deep Sleep mode	Deep Sleep mode Control: <table><tr><th>A[1:0]</th><th>Description</th></tr><tr><td>00</td><td>Normal Mode [POR]</td></tr><tr><td>11</td><td>Enter Deep Sleep Mode</td></tr></table> After this command initiated, the chip will enter Deep Sleep Mode, BUSY pad will keep output high. Remark: To Exit Deep Sleep mode, User required to send HWRESET to the driver	A[1:0]	Description	00	Normal Mode [POR]	11	Enter Deep Sleep Mode																		
A[1:0]	Description																																			
00	Normal Mode [POR]																																			
11	Enter Deep Sleep Mode																																			
0	1		0	0	0	0	0	0	A ₁	A ₀																										

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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	11	0	0	0	1	0	0	0	1	Data Entry mode setting	Define data entry sequence A[2:0] = 011 [POR] A [1:0] = ID[1:0] Address automatic increment / decrement setting The setting of incrementing or decrementing of the address counter can be made independently in each upper and lower bit of the address. 00 –Y decrement, X decrement, 01 –Y decrement, X increment, 10 –Y increment, X decrement, 11 –Y increment, X increment [POR] A[2] = AM Set the direction in which the address counter is updated automatically after data are written to the RAM. AM= 0, the address counter is updated in the X direction. [POR] AM = 1, the address counter is updated in the Y direction.
0	1		0	0	0	0	0	A ₂	A ₁	A ₀		
0	0	12	0	0	0	1	0	0	1	0	SW RESET	It resets the commands and parameters to their S/W Reset default values except R10h-Deep Sleep Mode During operation, BUSY pad will output high. Note: RAM are unaffected by this command.
0	0	14	0	0	0	1	0	1	0	0	HV Ready Detection	HV ready detection A[6:0] = 00h [POR] The command required CLKEN=1 and ANALOGEN=1. Refer to Register 0x22 for detail. After this command initiated, HV Ready detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).
0	1		0	A ₆	A ₅	A ₄	0	A ₂	A ₁	A ₀		A[6:4]=n for cool down duration: 10ms x (n+1) A[2:0]=m for number of Cool Down Loop to detect. The max HV ready duration is 10ms x (n+1) x (m) HV ready detection will be trigger after each cool down time. The detection will be completed when HV is ready. For 1 shot HV ready detection, A[7:0] can be set as 00h.

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Command Table																										
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description														
0	0	15	0	0	0	1	0	1	0	1	VCI Detection	VCI Detection A[2:0] = 100 [POR] , Detect level at 2.3V A[2:0] : VCI level Detect <table><tr><td>A[2:0]</td><td>VCI level</td></tr><tr><td>011</td><td>2.2V</td></tr><tr><td>100</td><td>2.3V</td></tr><tr><td>101</td><td>2.4V</td></tr><tr><td>110</td><td>2.5V</td></tr><tr><td>111</td><td>2.6V</td></tr><tr><td>Other</td><td>NA</td></tr></table> The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. After this command initiated, VCI detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).	A[2:0]	VCI level	011	2.2V	100	2.3V	101	2.4V	110	2.5V	111	2.6V	Other	NA
A[2:0]	VCI level																									
011	2.2V																									
100	2.3V																									
101	2.4V																									
110	2.5V																									
111	2.6V																									
Other	NA																									
0	1		0	0	0	0	0	A ₂	A ₁	A ₀																
0	0	18	0	0	0	1	1	0	0	0	Temperature Sensor Control	Temperature Sensor Selection A[7:0] = 48h [POR], external temperature sensor A[7:0] = 80h internal temperature sensor														
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																
0	0	1A	0	0	0	1	1	0	1	0	Temperature Sensor Control (Write to temperature register)	Write to temperature register. A[11:0] = 7FFh [POR]														
0	1		A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄																
0	1		A ₃	A ₂	A ₁	A ₀	0	0	0	0																
0	0	1B	0	0	0	1	1	0	1	1	Temperature Sensor Control (Read from temperature register)	Read from temperature register.														
1	1		A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄																
1	1		A ₃	A ₂	A ₁	A ₀	0	0	0	0																

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Command Table																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description												
0	0	1C	0	0	0	1	1	1	0	0	Temperature Sensor Control (Write Command to External temperature sensor)	Write Command to External temperature sensor. A[7:0] = 00h [POR], B[7:0] = 00h [POR], C[7:0] = 00h [POR], A[7:6] <table border="1"><tr><td>A[7:6]</td><td>Select no of byte to be sent</td></tr><tr><td>00</td><td>Address + pointer</td></tr><tr><td>01</td><td>Address + pointer + 1st parameter</td></tr><tr><td>10</td><td>Address + pointer + 1st parameter + 2nd pointer</td></tr><tr><td>11</td><td>Address</td></tr></table> A[5:0] – Pointer Setting B[7:0] – 1 st parameter C[7:0] – 2 nd parameter The command required CLKEN=1. Refer to Register 0x22 for detail. After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation.	A[7:6]	Select no of byte to be sent	00	Address + pointer	01	Address + pointer + 1st parameter	10	Address + pointer + 1st parameter + 2nd pointer	11	Address		
A[7:6]	Select no of byte to be sent																							
00	Address + pointer																							
01	Address + pointer + 1st parameter																							
10	Address + pointer + 1st parameter + 2nd pointer																							
11	Address																							
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀														
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀														
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀														
0	0	20	0	0	1	0	0	0	0	0	Master Activation	Activate Display Update Sequence The Display Update Sequence Option is located at R22h. BUSY pad will output high during operation. User should not interrupt this operation to avoid corruption of panel images.												
0	0	21	0	0	1	0	0	0	0	1	Display Update Control 1	RAM content option for Display Update A[7:0] = 00h [POR] A[7:4] Red RAM option <table border="1"><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table> A[3:0] BW RAM option <table border="1"><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table>	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀														

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Command Table												Command	Description
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0			
0	0	24	0	0	1	0	0	1	0	0	Write RAM (Black White) / RAM 0x24	After this command, data entries will be written into the BW RAM until another command is written. Address pointers will advance accordingly For White pixel: Content of Write RAM(BW) = 1 For Black pixel: Content of Write RAM(BW) = 0	
0	0	25	0	0	1	0	0	1	0	1	Write RAM (Dithering)	After this command, data entries will be written into the dithering engine.	
0	0	26	0	0	1	0	0	1	1	0	Write RAM (RED) / RAM 0x26	After this command, data entries will be written into the RED RAM until another command is written. Address pointers will advance accordingly. For Red pixel: Content of Write RAM(RED) = 1 For non-Red pixel [Black or White]: Content of Write RAM(RED) = 0	
0	0	27	0	0	1	0	0	1	1	1	Read RAM	After this command, data read on the MCU bus will fetch data from RAM [According to parameter of Register 41h to select reading RAM(BW) / RAM(RED)], until another command is written. Address pointers will advance accordingly. The 1 st byte of data read is dummy data.	
0	0	28	0	0	1	0	1	0	0	0	VCOM Sense	Enter VCOM sensing conditions and hold for duration defined in 29h before reading VCOM value. The sensed VCOM voltage is stored in register The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. BUSY pad will output high during operation.	
0	0	29	0	0	1	0	1	0	0	1	VCOM Sense Duration	Stabling time between entering VCOM sensing mode and reading acquired. A[6]=1, Normal Mode A[6]=0, Reserve A[3:0] = 09h, duration = 10s. VCOM sense duration = Setting + 1 Seconds	
0	1		0	A ₆	0	0	A ₃	A ₂	A ₁	A ₀			

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Command Table																																																																																														
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																																																																		
0	0	2A	0	0	1	0	1	0	1	0	Program VCOM OTP	Program VCOM register into OTP The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.																																																																																		
0	0	2B	0	0	1	0	1	0	1	1	Write Register for VCOM Control	This command is used to reduce glitch when ACVCOM toggle. Two data bytes D04h and D63h should be set for this command.																																																																																		
0	1		0	0	0	0	0	1	0	0																																																																																				
0	1		0	1	1	0	0	0	1	1																																																																																				
0	0	2C	0	0	1	0	1	1	0	0	Write VCOM register	Write VCOM register from MCU interface A[7:0] = 00h [POR]																																																																																		
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		<table><tr><th>A[7:0]</th><th>VCOM</th><th>A[7:0]</th><th>VCOM</th></tr><tr><td>08h</td><td>-0.2</td><td>58h</td><td>-2.2</td></tr><tr><td>0Ch</td><td>-0.3</td><td>5Ch</td><td>-2.3</td></tr><tr><td>10h</td><td>-0.4</td><td>60h</td><td>-2.4</td></tr><tr><td>14h</td><td>-0.5</td><td>64h</td><td>-2.5</td></tr><tr><td>18h</td><td>-0.6</td><td>68h</td><td>-2.6</td></tr><tr><td>1Ch</td><td>-0.7</td><td>6Ch</td><td>-2.7</td></tr><tr><td>20h</td><td>-0.8</td><td>70h</td><td>-2.8</td></tr><tr><td>24h</td><td>-0.9</td><td>74h</td><td>-2.9</td></tr><tr><td>28h</td><td>-1</td><td>78h</td><td>-3</td></tr><tr><td>2Ch</td><td>-1.1</td><td>7Ch</td><td>-3.1</td></tr><tr><td>30h</td><td>-1.2</td><td>80h</td><td>-3.2</td></tr><tr><td>34h</td><td>-1.3</td><td>84h</td><td>-3.2</td></tr><tr><td>38h</td><td>-1.4</td><td>88h</td><td>-3.4</td></tr><tr><td>3Ch</td><td>-1.5</td><td>8Ch</td><td>-3.5</td></tr><tr><td>40h</td><td>-1.6</td><td>90h</td><td>-3.6</td></tr><tr><td>44h</td><td>-1.7</td><td>94h</td><td>-3.7</td></tr><tr><td>48h</td><td>-1.8</td><td>98h</td><td>-3.8</td></tr><tr><td>4Ch</td><td>-1.9</td><td>9Ch</td><td>-3.9</td></tr><tr><td>50h</td><td>-2</td><td>A0h</td><td>-4</td></tr><tr><td>54h</td><td>-2.1</td><td>Others</td><td>Reserved</td></tr></table>	A[7:0]	VCOM	A[7:0]	VCOM	08h	-0.2	58h	-2.2	0Ch	-0.3	5Ch	-2.3	10h	-0.4	60h	-2.4	14h	-0.5	64h	-2.5	18h	-0.6	68h	-2.6	1Ch	-0.7	6Ch	-2.7	20h	-0.8	70h	-2.8	24h	-0.9	74h	-2.9	28h	-1	78h	-3	2Ch	-1.1	7Ch	-3.1	30h	-1.2	80h	-3.2	34h	-1.3	84h	-3.2	38h	-1.4	88h	-3.4	3Ch	-1.5	8Ch	-3.5	40h	-1.6	90h	-3.6	44h	-1.7	94h	-3.7	48h	-1.8	98h	-3.8	4Ch	-1.9	9Ch	-3.9	50h	-2	A0h	-4	54h	-2.1
A[7:0]	VCOM	A[7:0]	VCOM																																																																																											
08h	-0.2	58h	-2.2																																																																																											
0Ch	-0.3	5Ch	-2.3																																																																																											
10h	-0.4	60h	-2.4																																																																																											
14h	-0.5	64h	-2.5																																																																																											
18h	-0.6	68h	-2.6																																																																																											
1Ch	-0.7	6Ch	-2.7																																																																																											
20h	-0.8	70h	-2.8																																																																																											
24h	-0.9	74h	-2.9																																																																																											
28h	-1	78h	-3																																																																																											
2Ch	-1.1	7Ch	-3.1																																																																																											
30h	-1.2	80h	-3.2																																																																																											
34h	-1.3	84h	-3.2																																																																																											
38h	-1.4	88h	-3.4																																																																																											
3Ch	-1.5	8Ch	-3.5																																																																																											
40h	-1.6	90h	-3.6																																																																																											
44h	-1.7	94h	-3.7																																																																																											
48h	-1.8	98h	-3.8																																																																																											
4Ch	-1.9	9Ch	-3.9																																																																																											
50h	-2	A0h	-4																																																																																											
54h	-2.1	Others	Reserved																																																																																											
0	0	2D	0	0	1	0	1	1	0	1	OTP Register Read for Display Option	Read Register for Display Option:																																																																																		
1	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		A[7:0]: VCOM OTP Selection (Command 0x37, Byte A)																																																																																		
1	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		B[7:0]: VCOM Register (Command 0x2C)																																																																																		
1	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		C[7:0]~G[7:0]: Display Mode (Command 0x37, Byte B to Byte F) [5 bytes]																																																																																		
1	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		H[7:0]~K[7:0]: Waveform Version (Command 0x37, Byte G to Byte J) [4 bytes]																																																																																		
1	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀																																																																																				
1	1		F ₇	F ₆	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀																																																																																				
1	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀																																																																																				
1	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀																																																																																				
1	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀																																																																																				
1	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀																																																																																				
1	1		K ₇	K ₆	K ₅	K ₄	K ₃	K ₂	K ₁	K ₀																																																																																				

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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	2E	0	0	1	0	1	1	1	0	User ID Read	Read 10 Byte User ID stored in OTP: A[7:0]~J[7:0]: UserID (R38, Byte A and Byte J) [10 bytes]
1	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
1	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		
1	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		
1	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		
1	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		
1	1		F ₇	F ₆	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀		
1	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀		
1	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀		
1	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀		
1	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀		
0	0	2F	0	0	1	0	1	1	1	1	Status Bit Read	Read IC status Bit [POR 0x01] A[5]: HV Ready Detection flag [POR=0] 0: Ready 1: Not Ready A[4]: VCI Detection flag [POR=0] 0: Normal 1: VCI lower than the Detect level A[3]: [POR=0] A[2]: Busy flag [POR=0] 0: Normal 1: BUSY A[1:0]: Chip ID [POR=01] Remark: A[5] and A[4] status are not valid after RESET, they need to be initiated by command 0x14 and command 0x15 respectively.
1	1		0	0	A ₅	A ₄	0	0	A ₁	A ₀		
0	0	30	0	0	1	1	0	0	0	0	Program WS OTP	Program OTP of Waveform Setting The contents should be written into RAM before sending this command. The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	31	0	0	1	1	0	0	0	1	Load WS OTP	Load OTP of Waveform Setting The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	32	0	0	1	1	0	0	1	0	Write LUT register	Write LUT register from MCU interface [105 bytes], which contains the content of VS [nX-LUT], TP #[nX], RP#[n].
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		
0	1		:	:	:	:	:	:	:	:		
0	1		:	:	:	:	:	:	:	:		

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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	34	0	0	1	1	0	1	0	0	CRC calculation	CRC calculation command BUSY pad will output high during operation.
0	0	35	0	0	1	1	0	1	0	1	CRC Status Read	CRC Status Read A[15:0] is the CRC read out value
1	1		A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈		
1	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
0	0	36	0	0	1	1	0	1	1	0	Program OTP selection	Program OTP Selection according to the OTP Selection Control [R37h and R38h] The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	37	0	0	1	1	0	1	1	1	Write Register for Display Option	Write Register for Display Option B[7:0] Display Mode for WS[7:0] C[7:0] Display Mode for WS[15:8] D[7:0] Display Mode for WS[23:16] E[7:0] Display Mode for WS[31:24] F[3:0] Display Mode for WS[35:32] 0: Display Mode 1 1: Display Mode 2 F[6]: RAM Ping-Pong for Display Mode 2 1: RAM ping-pong enable 0: RAM ping-pong disable G[7:0]~J[7:0] module ID waveform version. Remarks: 1) A[7:0]~J[7:0] can be stored in OTP 2) RAM ping-pong function is not support for Display Mode 1
0	1		0	0	0	0	0	0	0	0		
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		
0	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		
0	1		0	F ₆	0	0	F ₃	F ₂	F ₁	F ₀		
0	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀		
0	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀		
0	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀		
0	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀		
0	0	38	0	0	1	1	1	0	0	0	Write Register for User ID	Write Register for User ID A[7:0]~J[7:0]: UserID [10 bytes] Remarks: A[7:0]~J[7:0] can be stored in OTP
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀		
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀		
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀		
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀		
0	1		E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀		
0	1		F ₇	F ₆	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀		
0	1		G ₇	G ₆	G ₅	G ₄	G ₃	G ₂	G ₁	G ₀		
0	1		H ₇	H ₆	H ₅	H ₄	H ₃	H ₂	H ₁	H ₀		
0	1		I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀		
0	1		J ₇	J ₆	J ₅	J ₄	J ₃	J ₂	J ₁	J ₀		

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Command Table																																										
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																														
0	0	39	0	0	1	1	1	0	0	1	OTP program mode	OTP program mode A[1:0] = 00: Normal Mode [POR] A[1:0] = 11: Internal generated OTP programming voltage Remark: User is required to EXACTLY follow the reference code sequences																														
0	1		0	0	0	0	0	0	A ₁	A ₀																																
0	0	3A	0	0	1	1	1	0	1	0	Reserved	Reserved																														
0	0	3B	0	0	1	1	1	0	1	1	Reserved	Reserved																														
0	0	3C	0	0	1	1	1	1	0	0	Border Waveform Control	Select border waveform for VBD A[7:0] = C0h [POR], set VBD as HiZ. A [7:6] :Select VBD option <table><tr><td>A[7:6]</td><td>Select VBD as</td></tr><tr><td>00</td><td>GS Transition, Defined in A[1:0]</td></tr><tr><td>01</td><td>Fix Level, Defined in A[5:4]</td></tr><tr><td>10</td><td>VCOM</td></tr><tr><td>11[POR]</td><td>HiZ</td></tr></table> A [5:4] Fix Level Setting for VBD <table><tr><td>A[5:4]</td><td>VBD level</td></tr><tr><td>00[POR]</td><td>VSS</td></tr><tr><td>01</td><td>VSH1</td></tr><tr><td>10</td><td>VSL</td></tr><tr><td>11</td><td>VSH2</td></tr></table> A [1:0] GS Transition setting for VBD <table><tr><td>A[1:0]</td><td>VBD Transition</td></tr><tr><td>00[POR]</td><td>LUT0</td></tr><tr><td>01</td><td>LUT1</td></tr><tr><td>10</td><td>LUT2</td></tr><tr><td>11</td><td>LUT3</td></tr></table>	A[7:6]	Select VBD as	00	GS Transition, Defined in A[1:0]	01	Fix Level, Defined in A[5:4]	10	VCOM	11[POR]	HiZ	A[5:4]	VBD level	00[POR]	VSS	01	VSH1	10	VSL	11	VSH2	A[1:0]	VBD Transition	00[POR]	LUT0	01	LUT1	10	LUT2	11	LUT3
A[7:6]	Select VBD as																																									
00	GS Transition, Defined in A[1:0]																																									
01	Fix Level, Defined in A[5:4]																																									
10	VCOM																																									
11[POR]	HiZ																																									
A[5:4]	VBD level																																									
00[POR]	VSS																																									
01	VSH1																																									
10	VSL																																									
11	VSH2																																									
A[1:0]	VBD Transition																																									
00[POR]	LUT0																																									
01	LUT1																																									
10	LUT2																																									
11	LUT3																																									
0	1		A ₇	A ₆	A ₅	A ₄	0	0	A ₁	A ₀																																
0	0	41	0	1	0	0	0	0	0	1	Read RAM Option	Read RAM Option A[0]= 0 [POR] 0: Read RAM corresponding to RAM 0x24 1: Read RAM corresponding to RAM 0x26																														
0	1		0	0	0	0	0	0	0	A ₀																																

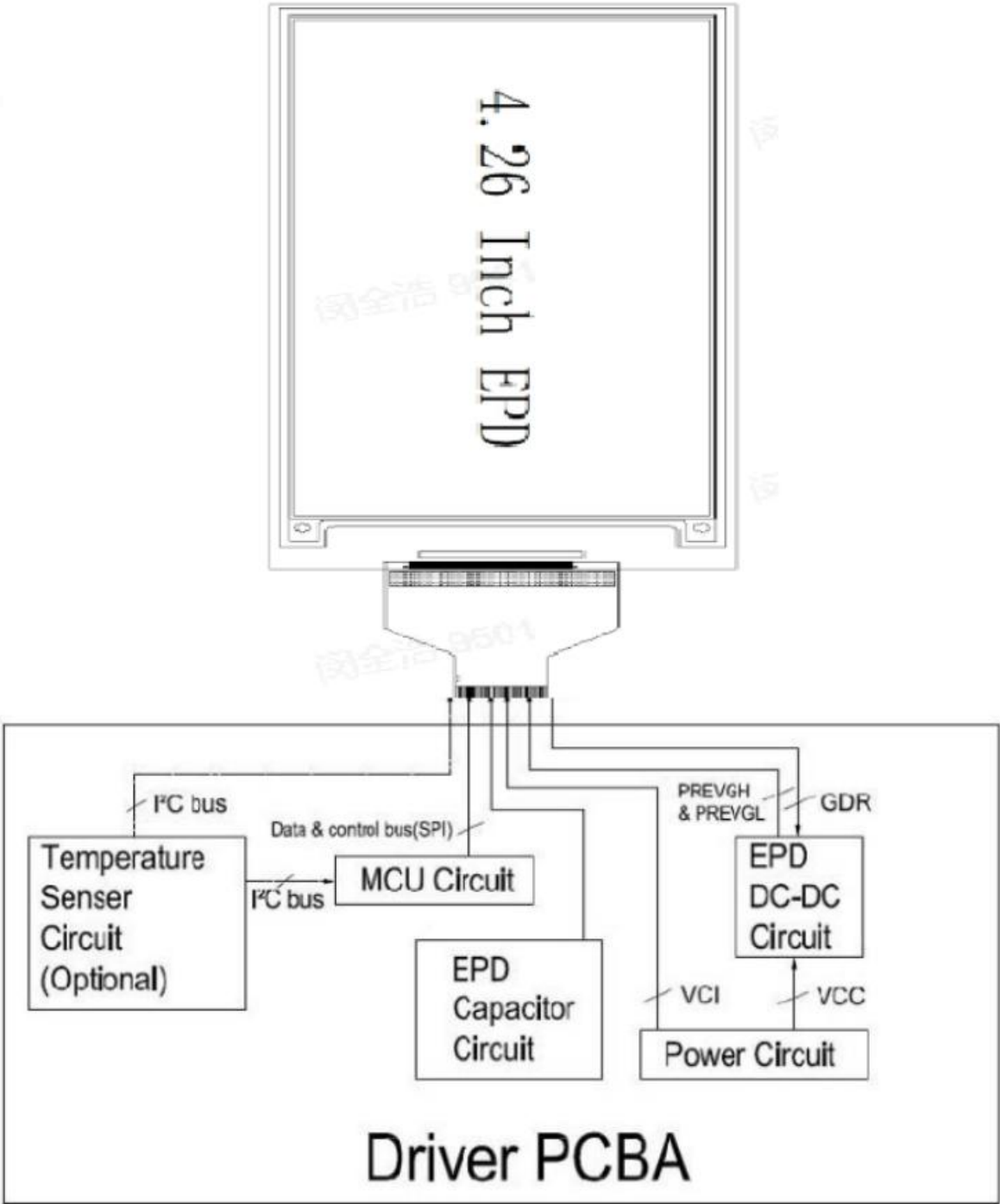
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Command Table																																																				
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																								
0	0	44	0	1	0	0	0	1	0	0	Set RAM X - address Start / End position	Specify the start/end positions of the window address in the X direction by an address unit for RAM A[9:0]: XSA[9:0], XStart, POR = 000h B[9:0]: XEA[9:0], XEnd, POR = 3BFh																																								
0	1		A ₇	A ₈	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		-	-	-	-	-	-	A ₉	A ₈																																										
0	1		0	0	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																										
0	1		-	-	-	-	-	-	B ₉	B ₈																																										
0	0	45	0	1	0	0	0	1	0	1	Set RAM Y- address Start / End position	Specify the start/end positions of the window address in the Y direction by an address unit for RAM A[9:0]: YSA[9:0], YStart, POR = 000h B[9:0]: YEA[9:0], YEnd, POR = 2A7h																																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		-	-	-	-	-	-	A ₉	A ₈																																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																										
0	1		-	-	-	-	-	-	B ₉	B ₈																																										
0	0	46	0	1	0	0	0	1	1	0	Auto Write RED RAM for Regular Pattern	Auto Write RED RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Height, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><th>A[6:4]</th><th>Height</th><th>A[6:4]</th><th>Height</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><th>A[2:0]</th><th>Width</th><th>A[2:0]</th><th>Width</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> BUSY pad will output high during operation.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	680																																																	
A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	960																																																	
0	1		A ₇	A ₆	A ₅	A ₄	0	A ₂	A ₁	A ₀																																										

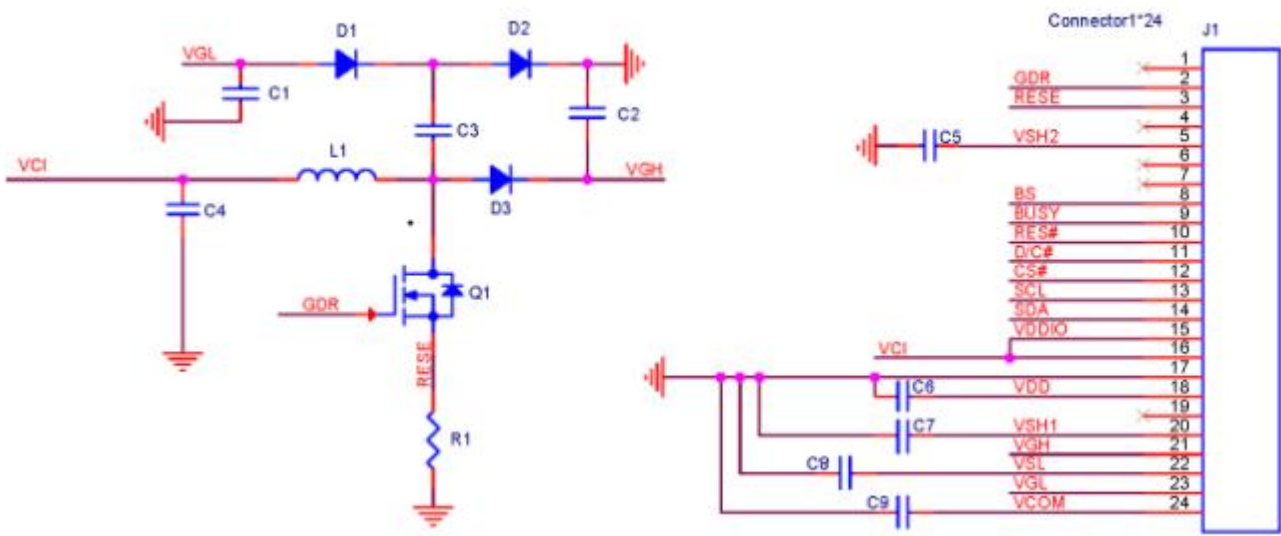
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Command Table												Command	Description																																							
0	0	47	0	1	0	0	0	1	1	1	Auto Write B/W RAM for Regular Pattern	Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Hieght, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><td>A[6:4]</td><td>Height</td><td>A[6:4]</td><td>Height</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><td>A[2:0]</td><td>Width</td><td>A[2:0]</td><td>Width</td></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> During operation, BUSY pad will output high.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
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A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	960																																																	
0	0	4D	0	1	0	0	1	1	0	1	Dithering engine Start/Stop	This command is used for start and stop dithering function, start commands should be sent before Command 0x25 and stop commands should be sent after the data bytes of command 0x25 <table><tr><td></td><td colspan="4">Data byte</td></tr><tr><td></td><td>A[7:0]</td><td>B[7:0]</td><td>C[7:0]</td><td>D[7:0]</td></tr><tr><td>Start Sierra Lite</td><td>80</td><td>00</td><td>78</td><td>00</td></tr><tr><td>Start Floyd-Steinberg</td><td>83</td><td>00</td><td>78</td><td>00</td></tr><tr><td>Stop dithering</td><td>00</td><td>00</td><td>78</td><td>00</td></tr></table>		Data byte					A[7:0]	B[7:0]	C[7:0]	D[7:0]	Start Sierra Lite	80	00	78	00	Start Floyd-Steinberg	83	00	78	00	Stop dithering	00	00	78	00															
	Data byte																																																			
	A[7:0]	B[7:0]	C[7:0]	D[7:0]																																																
Start Sierra Lite	80	00	78	00																																																
Start Floyd-Steinberg	83	00	78	00																																																
Stop dithering	00	00	78	00																																																
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀																																										
0	1		C ₇	C ₆	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀																																										
0	1		D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀																																										
0	0	4E	0	1	0	0	1	1	1	0	Set RAM X address counter	Make initial settings for the RAM X address in the address counter (AC) A[9:0]: 000h [POR].																																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		0	0	0	0	0	0	A ₉	A ₈																																										
0	0	4F	0	1	0	0	1	1	1	1	Set RAM Y address counter	Make initial settings for the RAM Y address in the address counter (AC) A[9:0]: 000h [POR].																																								
0	1		A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀																																										
0	1		0	0	0	0	0	0	A ₉	A ₈																																										
0	0	7F	0	1	1	1	1	1	1	1	NOP	This command is an empty command; it does not have any effect on the display module. However, it can be used to terminate Frame Memory Write or Read Commands.																																								

8. Block Diagram



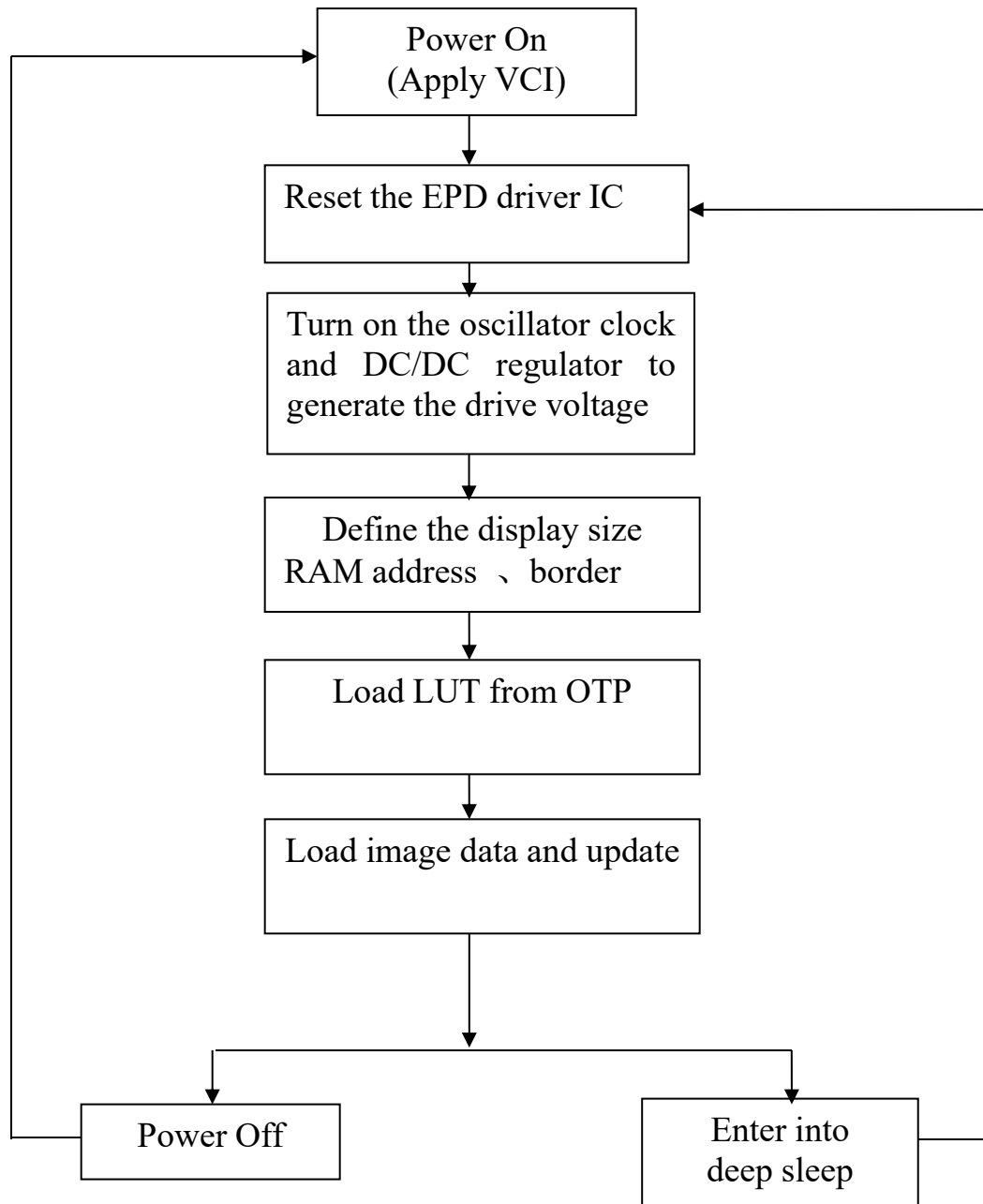
9. Typical Application Circuit with SPI Interface



Part Name	Value	Reference Part /Requirements for spare part
C4 C6	1uF	0603; X5R/X7R; Voltage Rating: 6V
C1 C2 C3 C5 C7 C8	4.7uF	0805; X5R/X7R; Voltage Rating: 25V
C9	1uF	0805; X7R; Voltage Rating: 25V
R1	2.2 Ohm	0805; 1%
D1 D2 D3	Diode	MBR0530 1) Reverse DC voltage $\geq 30V$ 2) $I_o \geq 500mA$ 3) Forward voltage $\leq 430mV$
Q1	NMOS	Si1304BDL/NX3008NBK 1) Drain-Source breakdown voltage $\geq 30V$ 2) $V_{gs(th)} = 0.9V$ (Typ), 1.3V (Max) 3) $R_{ds\ on} \leq 2.1\Omega$ @ $V_{gs} = 2.5V$
L1	47uH	CDRH2D18/ LDNP-470NC

10 Typical Operating Sequence

10.1 LUT from OTP Operation Flow



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10.2 OTP Operation Reference Program Code

ACTION	VALUE/DATA	COMMENT
POWER ON		
3v		
delay	10ms	
PIN CONFIG		
RES#	low	Hardware reset
delay	200us	
RES#	high	
delay	200us	
Read busy pin		Wait for busy low
Command 0x12		Software reset
Read busy pin		Wait for busy low
Command 0x0C	Data 0xAE 0xC7 0xC3 0xC0 0x80	Booster Soft-start Control
Command 0x01	Data 0xDF 0x01 0x02	Set display size and driver output control
Command 0x11	Data 0x01	Ram data entry mode
Command 0x44	Data 0x00 0x00 0x1F 0x03	Set Ram X address
Command 0x45	Data 0xDF 0x01 0x00 0x00	Set Ram Y address
Command 0x3C	Data 0x01	Set border
Command 0x18	Data 0x80	
LOAD IMAGE AND UPDATE		
Command 0x4E	Data 0x00 0x00	Set Ram X address counter
Command 0x4F	Data 0xDF 0x01	Set Ram Y address counter
Command 0x24	48000 bytes	Load BW image (800/8*480)
Command 0x22	Data 0xF7	Image update
Command 0x20		
Read busy pin		Wait for busy low
Command 0x10	Data 0X01	Enter deep sleep mode
POWER OFF		

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11. Reliability Test

NO	Test items	Test condition
1	Low Temp.Operation	0℃ for 240 hrs
2	High Temp. High Humidity Storage	60℃ / 80% RH for 240 hrs
3	Thermal Shock	1cycle: -25℃ / 30min ~ 60℃ / 30min for 100 cycles
4	HighTempLowhumidityStorage	60℃/35%RH for 240hrs
5	High Temp. High Humidity Operation	40℃/80%RH for 240hrs
6	ESD Gun	Air+/-4KV;Contact+/-2KV Contact+/-2KV(HBMC:100pF;R:1.5k ohm) Contact+/-200V(MMC:200pF;R:0 ohm) (Naked EPD display,including IC and FPC area)
7	Vibration test	Frequency: 10-500Hz Direction: X, Y, Z Duration: 1 hour in each direction
8	Dropping test	The test height is determined by the weight of the test object. Weight ≤20KG, drop height 1000mm Weight ≤50KG, drop height 500mm Weight ≤100KG, drop height 250mm

Note: 1. Stay white pattern for storage and non-operation test.
2. Operation is black→white pattern, the interval is 150s.

12. Quality Assurance

12.1 Environment

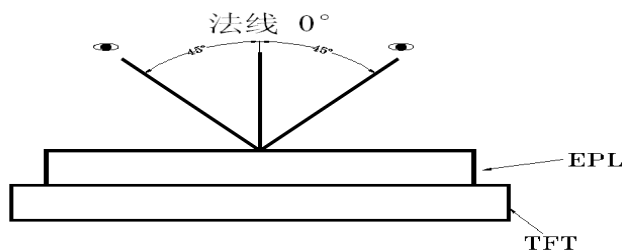
Temperature: $23 \pm 3^{\circ}\text{C}$

Humidity: $55 \pm 10\% \text{RH}$

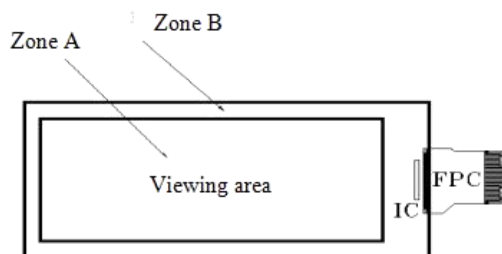
12.2 Illuminance

Brightness: $1200 \sim 1500 \text{LUX}$; distance: $20\text{-}30\text{CM}$; Angle: Relate 45° surround.

12.3 Inspect method

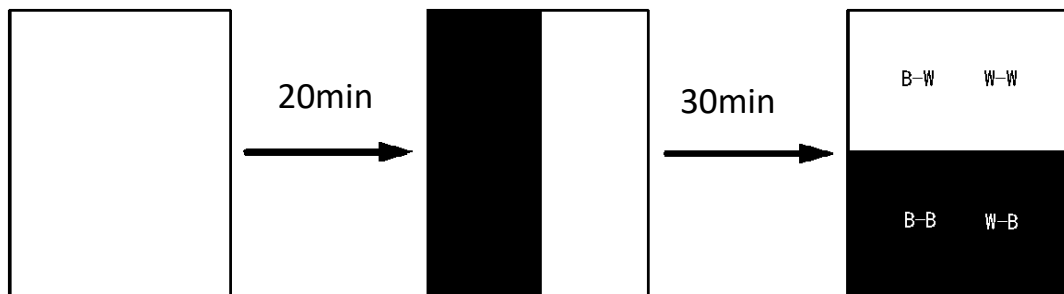


12.4 Display area



12.5 Ghosting test method

Two-color ghosting is measured with following transition from horizontal 2 scale pattern to vertical 2 scale pattern. The listed optical characteristics are only guaranteed under the controller & waveform provided by YRD.



1) Measurement Instruments: X-rite i1Pro

2) Ghosting formula:

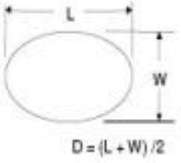
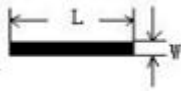
W ghosting: $\Delta L = \text{Max}(\Delta L(W-W, B-W)) - \text{Min}(\Delta L(W-W, B-W))$

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K ghosting: $\Delta L = \text{Max}(\Delta L(W-B, B-B)) - \text{Min}(\Delta L(W-B, B-B))$

12.6 Inspection standard

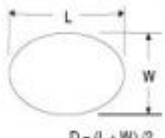
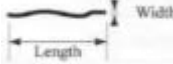
12.6.1 Electric inspection standard

NO.	Item	Standard	Defect level	Method	Scope
1	Display	Clear display、 Display complete、 Display uniform	MA	Visual inspection	Zone A
2	Black/White spots	 $D = (L + W) / 2$ $D \leq 0.3\text{mm}$, negligible $0.3\text{mm} < D \leq 0.5\text{mm}$, $N \leq 5$, Allowed $0.5\text{mm} < D$ Not Allow	MI	Visual Inspection card	Zone A
3	Black/White lines (No switch)	 $L \leq 1.0\text{mm}, W \leq 0.15\text{mm}$, negligible $1.0\text{mm} < L \leq 4.0\text{mm}$, $0.15\text{mm} < W \leq 0.5\text{mm}$, $N \leq 4$ allowable $L > 4.0\text{mm}, W > 0.5\text{mm}$ is not allowed	MI	Visual Inspection card	Zone A
4	Ghost image	Allowed in switching process	MI	Visual	Zone A
5	Flash dot / Multilateral	Flash points are allowed when switching screens Multilateral colors outside the frame are allowed for fixed screen time	MI	Visual/ Inspection card	Zone A
6	Segmented display	Selection segments are all displayed, and other segments are not displayed after the selection segment.	MA	Visual inspection card	Zone A
7	Short circuit/ Circuit break/ Abnormal Display	Not Allow	MA	Visual	Zone A

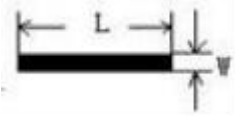
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8	Corner mura	$X > 1\text{mm}$, $Y > 1\text{mm}$, not allow 	MI	Visual/Ruler	Zone A
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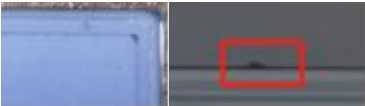
12.6.2 Appearance inspection standard

NO.	Item	Standard	Defect level	Method	Scope
1	B/W spots /Bubble/ Foreign bodies/ Dents	 $D \leq 0.3\text{mm}$, Allowed $0.3\text{mm} < D \leq 0.5\text{mm}$, $N \leq 5$ $D > 0.5\text{mm}$, Not Allow	MI	Visual/ inspection Card	Zone A
2	Chips/Scratch/ Edge crown	 $X \leq 3\text{mm}$, $Y \leq 0.5\text{mm}$ t= not counted , and without affecting the electrode , permissible  $X \leq 2\text{mm}$ or $Y \leq 2\text{mm}$ t= not counted.and without affecting the electrode , permissible  $W \leq 0.1\text{mm}$, $L \leq 5\text{mm}$, without affecting the electrode , $n \leq 2$	MI	Visual / Ruler	Zone A/B
3	TFT Cracks	 Not Allow	MA	Visual / Microscope	Zone A/B
4	Dirty/ foreign body	Allowed if can be removed	MI	Visual / clean cloth	Zone A / B

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5	FPC broken/ FPC oxidation / scratch	 Not Allow	MA	Visual / Microscope	Zone B
6	B/W Line	 $L \leq 1.0\text{mm}, W \leq 0.15\text{mm}$, negligible $1.0\text{mm} < L \leq 4.0\text{mm}$, $0.15\text{mm} < W \leq 0.5\text{mm}$, $N \leq 4$ allowable $L > 4.0\text{mm}, W > 0.5\text{mm}$ is not allowed	MI	Visual / Ruler	Zone A
7	TFT edge bulge /TFT chromatic aberration	 TFT edge bulge: $X \leq 3\text{mm}, Y \leq 0.3\text{mm}$ Allowed TFT chromatic aberration :Allowed	MI	Visual / Microscope	Zone A/B
8	Electrostatic point	$D \leq 0.35\text{mm}$, allow $0.35\text{mm} < D \leq 0.55\text{mm}$, $n \leq 5$ allow $D > 0.55\text{mm}$ is not allowed	MI	Visual /inspection card	Zone A
9	PCB damaged/ Poor welding/ Curl	PCB (Circuit area) damaged Not Allow Allow PCB Poor welding Not Allow PCB Curl $\leq 1\%$	MA	Visual / Ruler	Zone B
10	Edge glue height/ Edge glue bubble	Edge Adhesives $H \leq$ PS surface (Including protect film) Edge adhesives overflow onto protect film $\leq 1/2$ FPL to PS Margin width Length excluding Edge adhesives seep in $\leq 1/2$ FPL Margin width Length excluding Edge adhesives bubble: bubble Width $\leq 1/2$ Margin width; Length $\leq 5.0\text{mm}$. $n \leq 5$	MI	Visual / Ruler	Zone B
11	Protect film	Surface scratch but not effect protect function, Allow	MI	Visual	Zone A/B

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12	Silicon glue	Thickness≤PS surface(With protect film): Full cover the IC; Shape: The width on the FPC≤1.0mm (Front) The width on the FPC≤1.0mm (Back) smooth surface, No obvious raised.	MI	Visual/Ruler	Zone B
13	Warp degree (TFT substrate)	 $t \leq 1.5\text{mm}$	MI	Ruler	Zone B
14	Color difference in COM area (Silver point area)	Allowed	MI	Visual	Zone B
15	Corner and edge damage of FPL	Corner and edge damage $\leq 1/2$ border width 	MI	Visual / Ruler	Zone A

13. Packaging

14. Handling, Safety, and Environment Requirements

Warning

The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap.

Caution

The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. Disassembling the display module.

Disassembling the display module can cause permanent damage and invalidates the warranty agreements.

Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions.

Data sheet status	
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	
Product Environmental certification	
ROHS	
REMARK	
All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.	
Transport environment	
When the humidity of transportation environment is between 45%RH~70%RH, the product can be stored for 30 days, and the product can be stored for 10 days if it is lower or higher than this range	